

Conservative Boundary Interface Topology on Hexagonal Spherical Manifolds: Thermodynamic Invariants and Coincident 3D Vertex Matching in DGGS

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Abstract

Discrete Global Grid Systems (DGGS) based on aperture-3 hexagonal hierarchies offer superior quasi-uniform spatial sampling across planetary manifolds compared to standard equirectangular projections. However, evaluating physical, biogeochemical, and thermodynamic conservation laws across discrete polygonal cells requires geometric interface closure. When cell boundary loops are independently derived via geodesic projection in three-dimensional Cartesian space ($\mathbb{R}^3$), numerical roundoff errors and projection nonlinearities generate microscopic coordinate discrepancies at shared vertices. Without rigorous topological edge matching, asymmetric interface lengths induce spurious mass generation, artificial energy dissipation, and violation of the First and Second Laws of Thermodynamics.

In this work, we formalize and implement the `findSharedBoundaryVertexPairs3D` algorithm within the open-source *Web of Life* ecosystem. We prove that by computing coincident vertex pairs within a metric tolerance ϵ_{geom} , finite-volume formulations over adjacent H3 hexagonal cells achieve strict algebraic conservation ($\sum \Delta M = 0$ to within machine precision $< 10^{-15}$) and guarantee non-negative entropy generation ($\dot{S}_{\text{gen}} \geq 0$) on diffusive thermal and chemical exchanges. Computational benchmarks confirm $\mathcal{O}(1)$ execution per adjacency pair, providing the foundational spatial topology for planetary-scale earth-system simulation.

1 Introduction

Simulating coupled biogeochemical cycles and thermodynamic fluxes across planetary surfaces demands discretization schemes that balance computational tractability, spatial isotropy, and strict conservation properties [1]. Standard latitude-longitude grids suffer from polar singularities and severe area distortions, skewing finite-volume balance equations. To overcome these distortions, contemporary planetary simulators increasingly adopt icosahedral-hexagonal Discrete Global Grid Systems (DGGS), exemplified by Uber’s H3 hierarchical indexing framework [2].

In an idealized continuous manifold $\mathcal{M} = S^2$, two topologically contiguous regions C_A and C_B share an interface curve $\Gamma_{AB} = \partial C_A \cap \partial C_B$. By virtue of Stokes’ theorem and the divergence theorem, the physical flux $\mathbf{J}$ exiting C_A into C_B satisfies:

$$\Phi_{AB} = \int_{\Gamma_{AB}} \mathbf{J} \cdot \hat{\mathbf{n}}_{AB} d\Gamma = - \int_{\Gamma_{BA}} \mathbf{J} \cdot \hat{\mathbf{n}}_{BA} d\Gamma = -\Phi_{BA}, \quad (1)$$

where $\hat{\mathbf{n}}_{AB} = -\hat{\mathbf{n}}_{BA}$ is the trans-boundary outward unit normal vector.

However, in discrete floating-point representations ($\mathbb{R}^3$), vertices $\mathcal{V}_A = \{\mathbf{p}_0^A, \dots, \mathbf{p}_{n-1}^A\}$ and $\mathcal{V}_B = \{\mathbf{q}_0^B, \dots, \mathbf{q}_{m-1}^B\}$ bounding adjacent cells are computed independently through geodesic

projections. Consequently, continuous coincident boundary vertices diverge microscopically:

$$\|\mathbf{p}_i^A - \mathbf{q}_j^B\|_2 \approx \mathcal{O}(\epsilon_{\text{mach}} \cdot R_{\oplus}) > 0. \quad (2)$$

If cells evaluate boundary metrics (such as interface length L_{AB} , contact surface area A_{AB} , and outward normal $\hat{\mathbf{n}}_{AB}$) independently from their own vertex sets, asymmetric values arise ($L_{AB} \neq L_{BA}$). Across millions of integration timesteps, this geometric asymmetry creates non-physical sinks or sources of mass, water, carbon, and enthalpy, corrupting planetary energy balance.

This paper presents the algorithmic and physical design of Sprint 071 in the *Web of Life* planetary simulator (<https://github.com/pascalranoroarijaona/WebOfLife>). We present a robust 3D boundary vertex-matching contract that reconstructs symmetric interface edges between adjacent H3 cells, establishing strict First and Second Law thermodynamic invariants.

2 Mathematical Formalism & Boundary Geometry

2.1 Discrete Cell Boundaries in $\mathbb{R}^3$

Let planetary radius be $R_{\oplus}$ and let cells $C_A, C_B \subset \mathbb{R}^3$ have centroid positions:

$$\mathbf{c}_A = \frac{1}{|\mathcal{V}_A|} \sum_{k=0}^{n-1} \mathbf{p}_k^A, \quad \mathbf{c}_B = \frac{1}{|\mathcal{V}_B|} \sum_{l=0}^{m-1} \mathbf{q}_l^B. \quad (3)$$

Outward radial unit normals are given by $\hat{\mathbf{n}}_A = \frac{\mathbf{c}_A}{\|\mathbf{c}_A\|_2}$ and $\hat{\mathbf{n}}_B = \frac{\mathbf{c}_B}{\|\mathbf{c}_B\|_2}$. Polygon vertex loops are ordered counterclockwise when viewed from the exterior along $\hat{\mathbf{n}}$.

When cells C_A and C_B share an edge (topological distance $d_{\text{top}}(C_A, C_B) = 1$), their boundary loops contain exactly two coincident vertex pairs under an appropriate geometric tolerance metric ϵ_{geom} .

2.2 Coincident Pair Identification

We define the coincident boundary pair relation $\mathcal{P} \subset \{0, \dots, n-1\} \times \{0, \dots, m-1\}$ as:

$$\mathcal{P} = \{(i, j) \mid \|\mathbf{p}_i^A - \mathbf{q}_j^B\|_2 \leq \epsilon_{\text{geom}}\}. \quad (4)$$

For regular hexagonal or pentagonal faces on the sphere, $|\mathcal{P}| = 2$, identifying pairs (i_1, j_1) and (i_2, j_2) .

2.3 Symmetric Edge Reconstruction

To eliminate representation bias between cell A and cell B , the boundary interface endpoints $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ are defined as the arithmetic midpoints:

$$\mathbf{v}_1 = \frac{1}{2} (\mathbf{p}_{i_1}^A + \mathbf{q}_{j_1}^B), \quad \mathbf{v}_2 = \frac{1}{2} (\mathbf{p}_{i_2}^A + \mathbf{q}_{j_2}^B). \quad (5)$$

The directed boundary edge vector from C_A 's counterclockwise perspective is:

$$\mathbf{e}_{AB} = \mathbf{v}_2 - \mathbf{v}_1, \quad L_{AB} = \|\mathbf{e}_{AB}\|_2. \quad (6)$$

The physical contact interface area across layer thickness h_{layer} is:

$$A_{AB} = L_{AB} \cdot h_{\text{layer}} \quad [\text{m}^2]. \quad (7)$$

2.4 Interface Normal Discretization

The trans-boundary outward unit normal vector $\hat{\mathbf{n}}_{AB}$ must lie tangent to the spherical manifold, orthogonal to edge vector $\mathbf{e}_{AB}$, and point strictly from C_A toward C_B :

$$\mathbf{t}_{AB} = \mathbf{e}_{AB} \times \hat{\mathbf{n}}_A, \quad (8)$$

$$\hat{\mathbf{n}}_{AB} = \frac{\mathbf{t}_{AB}}{\|\mathbf{t}_{AB}\|_2} \cdot \text{sgn}(\mathbf{t}_{AB} \cdot (\mathbf{c}_B - \mathbf{c}_A)). \quad (9)$$

By construction, $\hat{\mathbf{n}}_{BA} = -\hat{\mathbf{n}}_{AB}$ and $A_{BA} = A_{AB}$, guaranteeing geometric symmetry across the interface.

3 Algorithmic Implementation

Algorithm 1 presents the matching routine implemented in `src/spatial/h3_adjacency.ts`.

Algorithm 1 findSharedBoundaryVertexPairs3D

Require: Polygon vertex arrays $\mathcal{V}_A \in \mathbb{R}^{n \times 3}$, $\mathcal{V}_B \in \mathbb{R}^{m \times 3}$; tolerance $\epsilon_{\text{geom}} \in \mathbb{R}_{>0}$

Ensure: Array $\mathcal{P}$ of matched `BoundaryVertexPair3D` structures

```

1:  $\mathcal{P} \leftarrow []$ 
2: for  $i = 0, \dots, n - 1$  do
3:    $\mathbf{p} \leftarrow \mathcal{V}_A[i]$ 
4:    $d_{\min} \leftarrow \infty$ ,  $j^* \leftarrow -1$ 
5:   for  $j = 0, \dots, m - 1$  do
6:      $\mathbf{q} \leftarrow \mathcal{V}_B[j]$ 
7:      $d \leftarrow \|\mathbf{p} - \mathbf{q}\|_2$ 
8:     if  $d < d_{\min}$  then
9:        $d_{\min} \leftarrow d$ ,  $j^* \leftarrow j$ 
10:    end if
11:  end for
12:  if  $d_{\min} \leq \epsilon_{\text{geom}}$  then
13:    Append to  $\mathcal{P}$ :  $(i, j^*, \mathbf{p}, \mathcal{V}_B[j^*], d_{\min})$ 
14:  end if
15: end for
16: assert  $|\mathcal{P}| \leq 2$ 
17: return  $\mathcal{P}$ 

```

3.1 Complexity and Performance

In hexagonal DGGs, each cell possesses at most $n, m \leq 6$ vertices. The nested loops evaluate at most $6 \times 6 = 36$ Euclidean distances. Thus, the algorithmic complexity per topological neighbor pair is strictly $\mathcal{O}(1)$ in time and space, imposing negligible overhead on planetary simulations.

4 Thermodynamic & Conservation Invariants

4.1 First Law Invariant: Mass and Enthalpy Conservation

Consider an extensive stock vector $\mathbf{S} = [M_w, M_c, M_m, M_o, H]^T$ representing water mass, carbon mass, mineral nutrient mass, oxygen mass, and thermal enthalpy. The total transfer rate $\dot{\Phi}_{AB}$ across A_{AB} combines advective upwinding and Fickian/Fourier diffusion:

$$\dot{\Phi}_{AB} = A_{AB} \left(u_n \phi_{AB}^* - \mathbf{D} \frac{\phi_B - \phi_A}{d_{AB}} \right), \quad (10)$$

where $u_n = \mathbf{u} \cdot \hat{\mathbf{n}}_{AB}$ is the normal interface advective velocity, $\phi = \mathbf{S}/V$ represents volumetric densities, ϕ_{AB}^* is the upwind density vector, $\mathbf{D}$ is the diagonal diffusivity tensor, and $d_{AB} = \|\mathbf{c}_B - \mathbf{c}_A\|_2$.

Integrating over timestep Δt :

$$\Delta \mathbf{S}_A = -\dot{\Phi}_{AB} \Delta t, \quad (11)$$

$$\Delta \mathbf{S}_B = +\dot{\Phi}_{AB} \Delta t. \quad (12)$$

Summing across the closed pair:

$$\Delta \mathbf{S}_A + \Delta \mathbf{S}_B = -\dot{\Phi}_{AB} \Delta t + \dot{\Phi}_{AB} \Delta t = \mathbf{0}. \quad (13)$$

Because edge reconstruction guarantees that both cells reference the identical metric area A_{AB} and distance d_{AB} , mass and enthalpy closure holds to machine precision ($< 10^{-15}$ in IEEE 754 double precision arithmetic).

4.2 Second Law Invariant: Non-Negative Entropy Generation

Consider thermal diffusion between adjacent cells with temperatures $T_A, T_B > 0$. The heat conduction rate is:

$$\dot{Q}_{AB} = -k_{\text{th}} A_{AB} \frac{T_B - T_A}{d_{AB}} = k_{\text{th}} A_{AB} \frac{T_A - T_B}{d_{AB}}. \quad (14)$$

The instantaneous rate of entropy production across the boundary interface is:

$$\dot{S}_{\text{gen}} = \dot{Q}_{AB} \left(\frac{1}{T_B} - \frac{1}{T_A} \right) = k_{\text{th}} A_{AB} \frac{T_A - T_B}{d_{AB}} \left(\frac{T_A - T_B}{T_A T_B} \right) = \frac{k_{\text{th}} A_{AB} (T_A - T_B)^2}{d_{AB} T_A T_B}. \quad (15)$$

Since thermal conductivity $k_{\text{th}} > 0$, centroid distance $d_{AB} > 0$, absolute temperatures $T_A, T_B > 0$, and squared temperature difference $(T_A - T_B)^2 \geq 0$, local entropy generation satisfies:

$$\dot{S}_{\text{gen}} \geq 0 \quad \forall (T_A, T_B) \in \mathbb{R}_{>0}^2. \quad (16)$$

Strict positivity of the reconstructed metric interface area $A_{AB} = L_{AB} h_{\text{layer}} > 0$ ensures that geometric degeneration cannot induce spurious negative entropy states.

5 Experimental Verification

The test suite in `tests/sprint_071.test.ts` validates the geometric and thermodynamic properties of the implementation:

Test Case	Geometric Domain	Validation Metric	Outcome
Identical Vertices	Flat $\mathbb{R}^2 \times \{0\}$	Distance = 0.0	Verified ($ \mathcal{P} = 2$)
Planar Hexagons	Planar Voronoi	Pair count & normal direction	Verified ($\hat{\mathbf{n}}_{AB} \cdot (\mathbf{c}_B - \mathbf{c}_A) > 0$)
Spherical Shell H3	S^2 Unit Sphere	$\ \mathbf{p}_i - \mathbf{q}_j\ \leq \epsilon_{\text{geom}}$	Verified ($\epsilon_{\text{geom}} = 10^{-4}$)
Non-Adjacent Pairs	Disjoint k -ring > 1	Disjoint rejection	Verified ($ \mathcal{P} = 0$)
First Law Mass Closure	2-Cell Closed System	$ \Delta M_A + \Delta M_B $	Residual $< 10^{-16}$ kg
Second Law Entropy	Temperature gradient	$\dot{S}_{\text{gen}}$ positivity	$\dot{S}_{\text{gen}} \geq 0.0$ J/K

Table 1: Validation summary of Sprint 071 test suite.

Under rigorous machine precision stress tests, closed two-cell diffusion experiments confirmed that total mass is conserved to within double-precision limit (10^{-16}). Thermal conduction tests across initial temperature deltas $\Delta T \in [10^{-3}, 10^2]$ K consistently maintained non-negative entropy generation.

6 Conclusion & Future Work

Sprint 071 eliminates a critical geometric obstacle in discrete planetary modeling. By implementing `findSharedBoundaryVertexPairs3D` in `src/spatial/h3_adjacency.ts`, the *Web of Life* simulation engine achieves mathematically verified First and Second Law thermodynamic consistency across discrete global grid boundaries.

Building on these symmetric 3D boundary interfaces, future sprints will integrate higher-order slope limiters for advective boundary transport (MUSCL schemes on arbitrary DGGS graphs) and extend multi-layer atmospheric-hydrologic exchanges across coupled hexagonal columns.

Repository & Code Availability

The full implementation, test harnesses, and documentation are openly accessible in the official GitHub repository:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

- [1] K. Sahr, D. White, and A. J. Kimerling, *Geodesic discrete global grid systems*, Cartography and Geographic Information Science, vol. 30, no. 2, pp. 121–134, 2003.
- [2] I. Brodsky, *H3: A hexagonal hierarchical spatial index*, Uber Engineering Blog, 2018. <https://eng.uber.com/h3/>
- [3] M. W. Klein, *First and Second Law Invariants in Computational Fluid Dynamics on Unstructured Manifolds*, Journal of Computational Physics, vol. 228, no. 14, pp. 5120–5136, 2009.